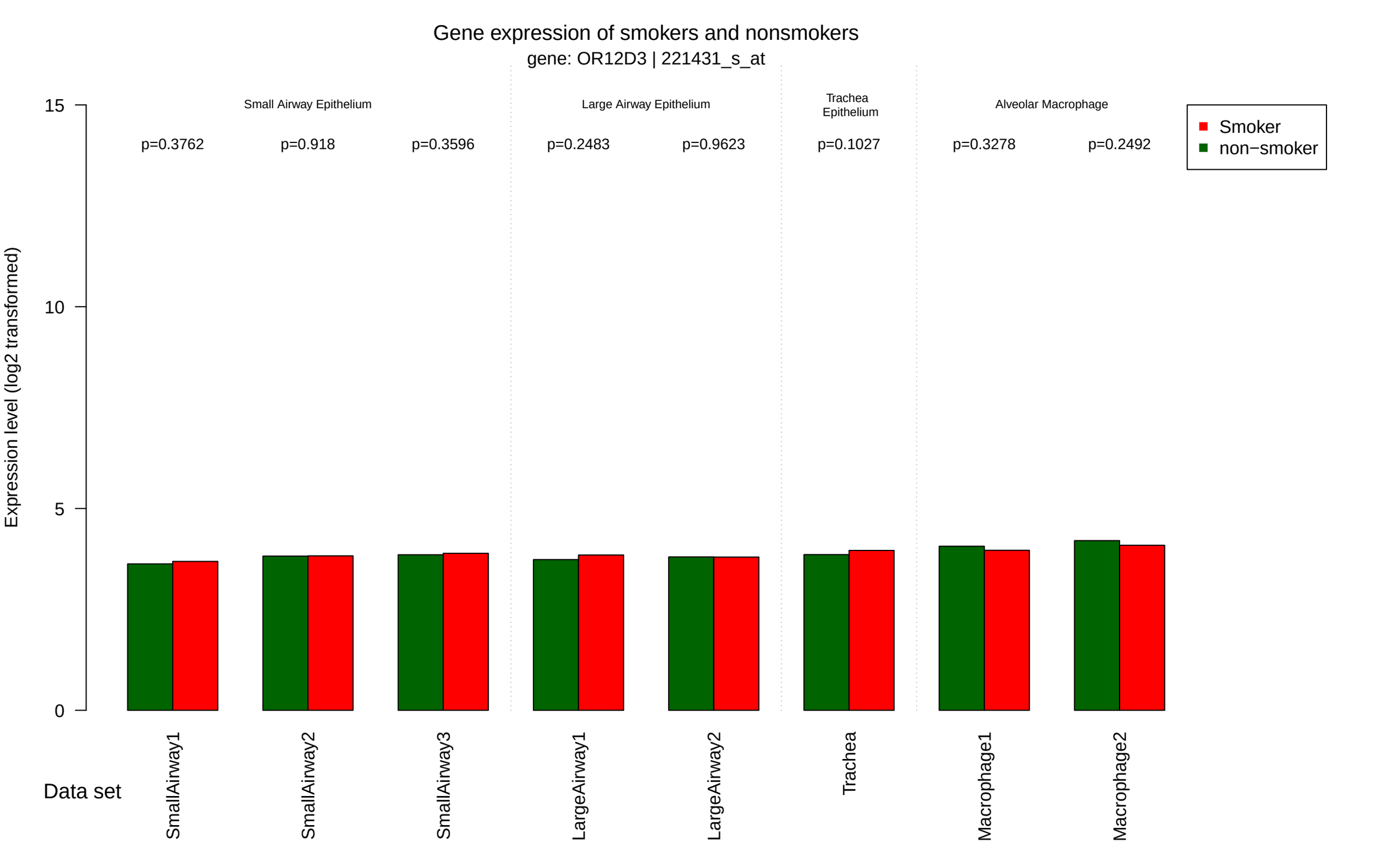
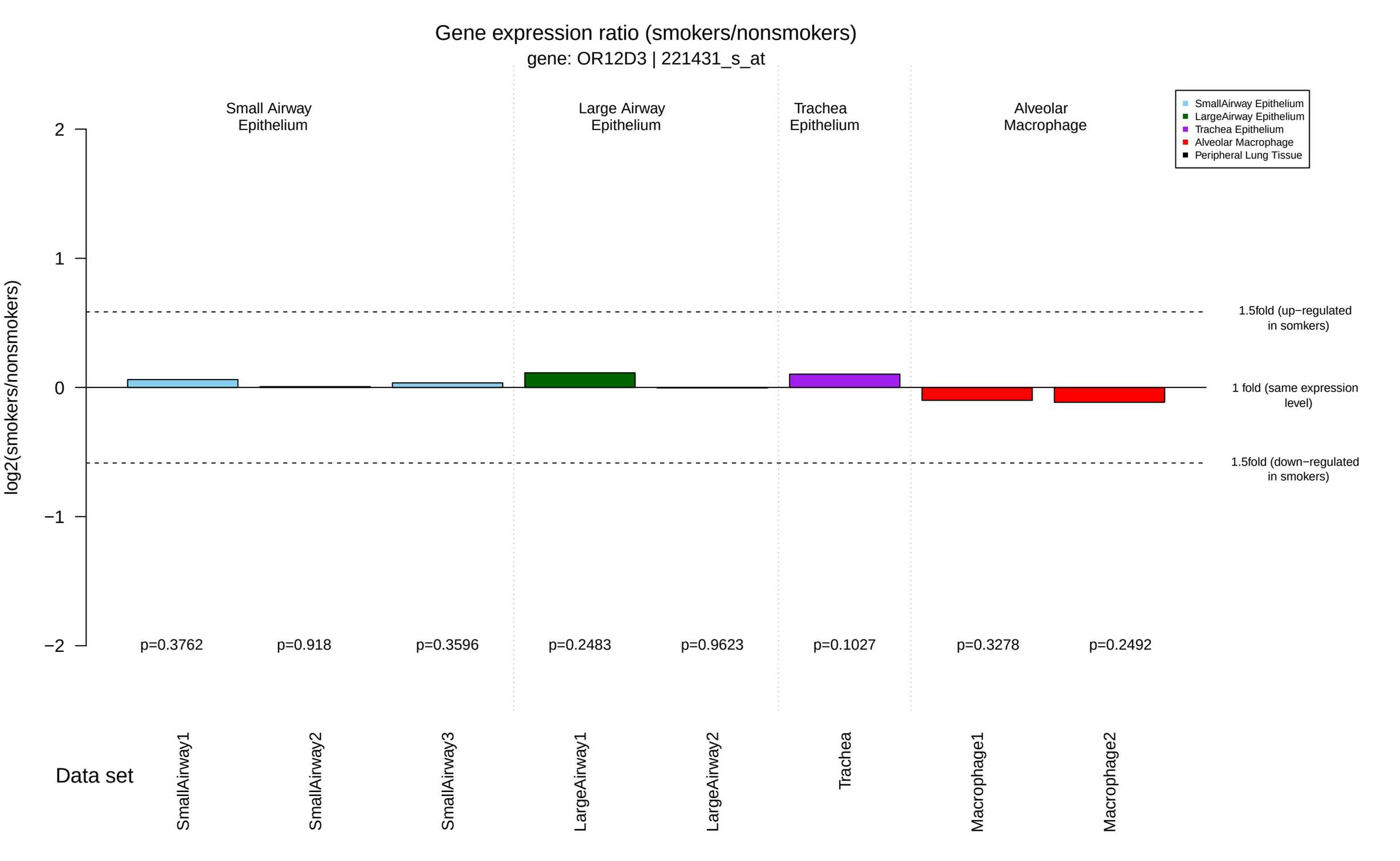
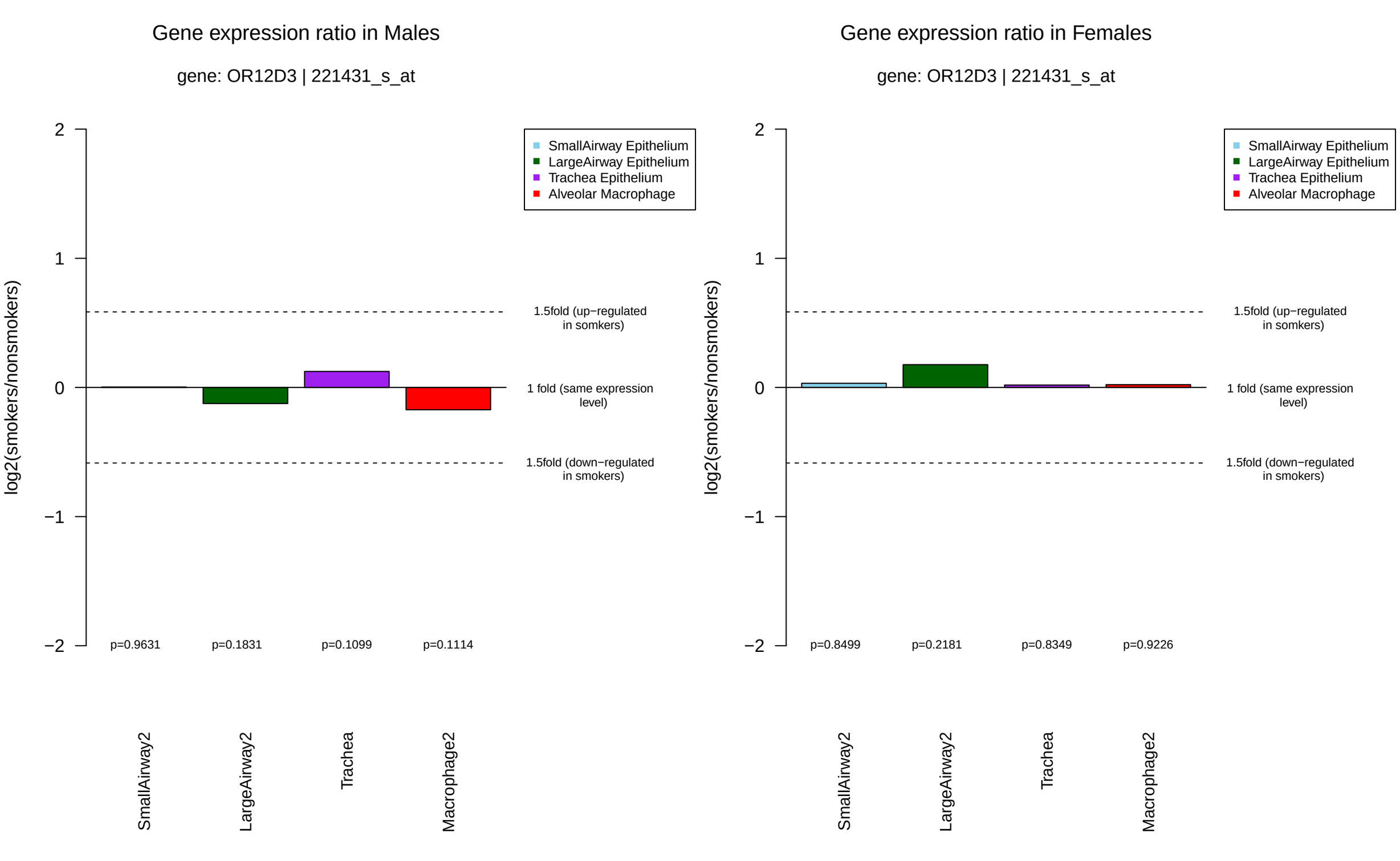
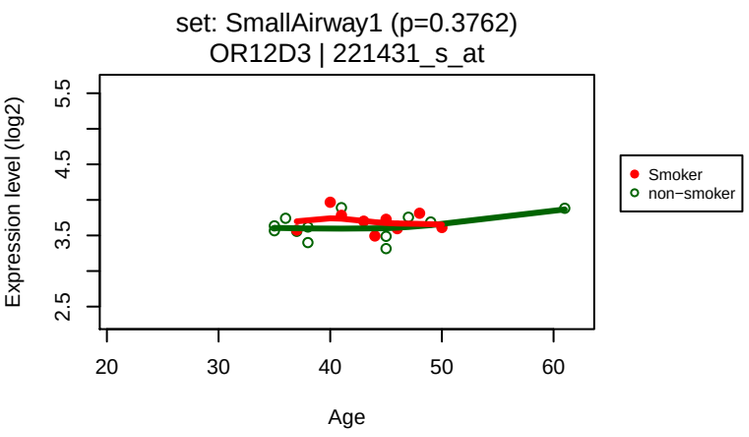
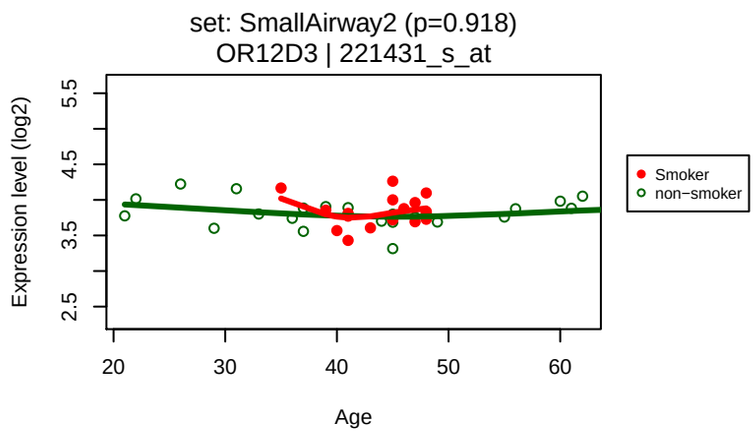
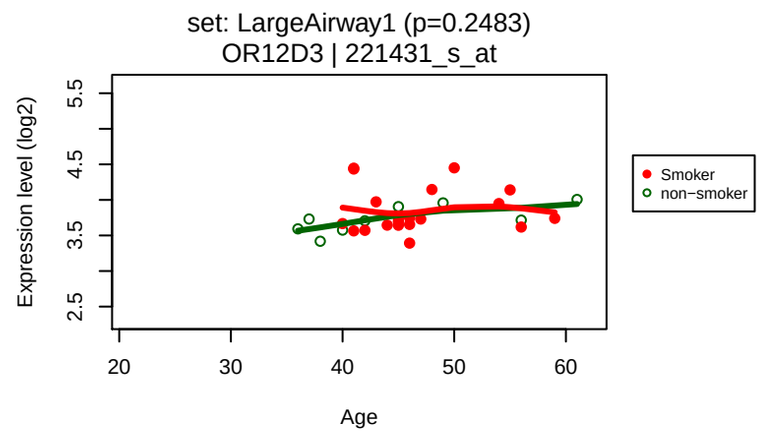

Gene expression of smokers and nonsmokers
gene: OR12D3 | 221431_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3762
p=0.918
p=0.3596
p=0.2483
p=0.9623
p=0.1027
p=0.3278
p=0.2492
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

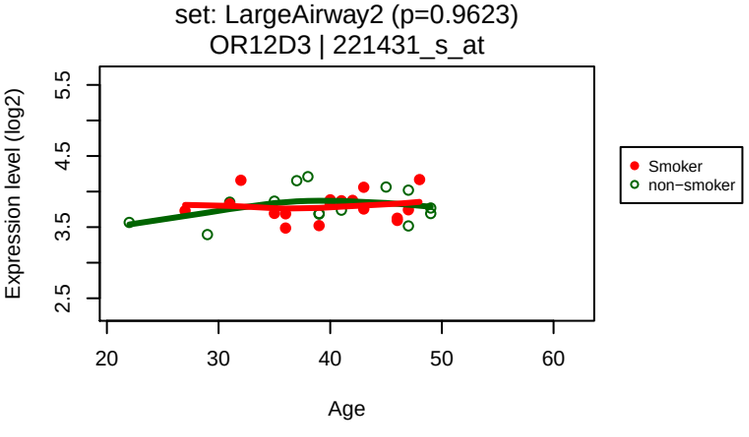

Gene expression ratio (smokers/nonsmokers)
gene: OR12D3 | 221431_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3762
p=0.918
p=0.3596
p=0.2483
p=0.9623
p=0.1027
p=0.3278
p=0.2492
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

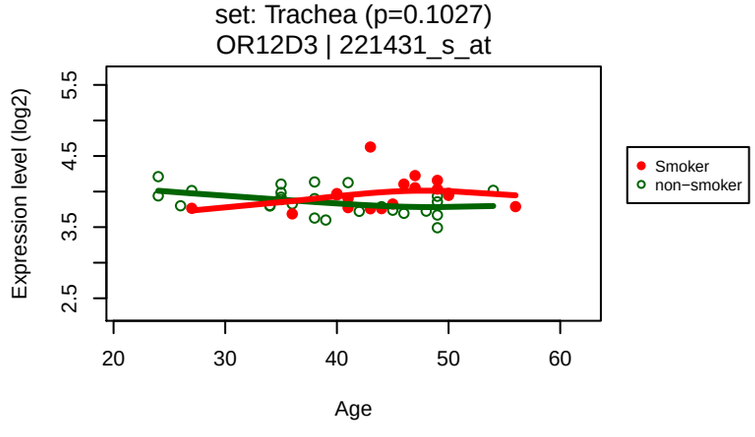

Gene expression ratio in Males
gene: OR12D3 | 221431_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9631
p=0.1831
p=0.1099
p=0.1114
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR12D3 | 221431_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8499
p=0.2181
p=0.8349
p=0.9226
SmallAirway2
LargeAirway2
Trachea
Macrophage2

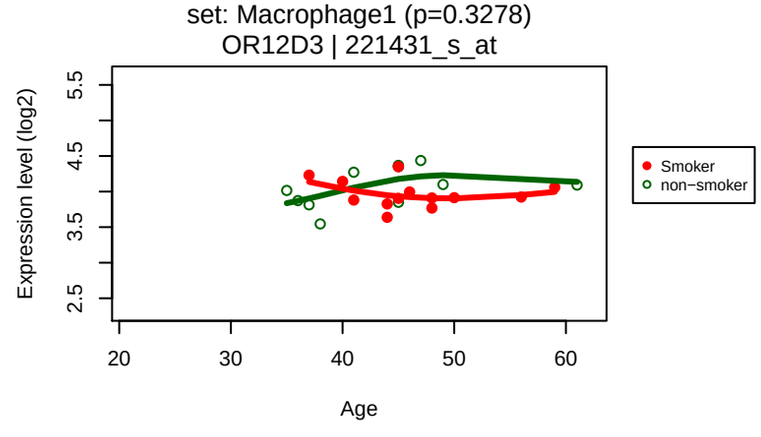

set: SmallAirway1 (p=0.3762)
OR12D3 | 221431_s_at
Expression level (log2)
Age
Smoker
non−smoker

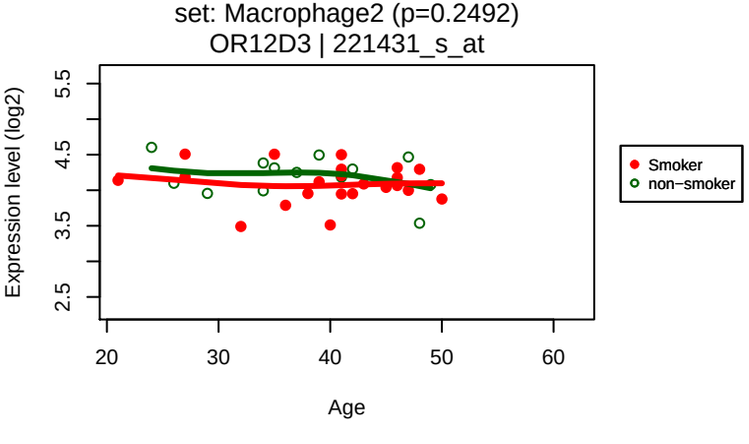

set: SmallAirway2 (p=0.918)
OR12D3 | 221431_s_at
Expression level (log2)
Age
Smoker
non−smoker

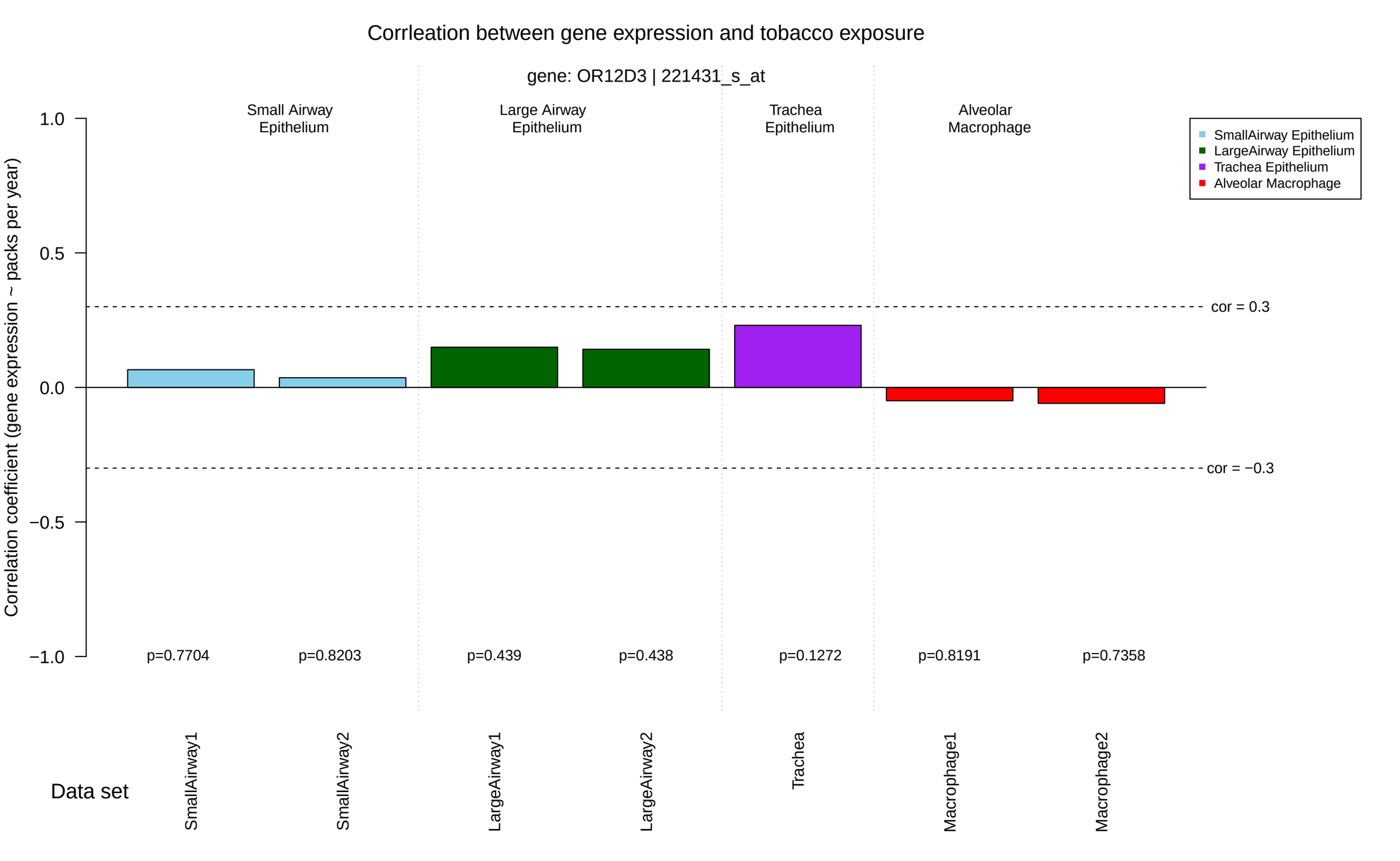

set: LargeAirway1 (p=0.2483)
OR12D3 | 221431_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9623)
OR12D3 | 221431_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1027)
OR12D3 | 221431_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3278)
OR12D3 | 221431_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2492)
OR12D3 | 221431_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR12D3 | 221431_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7704
p=0.8203
p=0.439
p=0.438
p=0.1272
p=0.8191
p=0.7358
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2